During this, K^+ ion gates are more permeable to K^+ even after reaching the threshold level as it closes slowly; hence called **Lazy gates**. The membrane potential returns to its original **resting state** when K^+ ion channels close completely. During hyperpolarization the Na^+ voltage gate remains closed (Figure 10.5).

Conduction Speed of a nerve impulse

The conduction speed of a nerve impulse depends on the diameter of axon. The greater the axon's diameter, the faster is the conduction. The **myelinated axon** conducts the impulse faster than the **non-myelinated axon**. The voltage-gated Na^+ and K^+ channels are concentrated at the **nodes of Ranvier**. As a result, the impulse jumps node to node, rather than travelling the entire length of the nerve fibre. This mechanism of conduction is called **Saltatory Conduction**. Nerve impulses travel at the speed of 1-300 m/s.

10.3.3 Synaptic transmission

The junction between two neurons is called a **Synapse** through which a nerve impulse is transmitted. The first neuron involved in the synapse forms the **pre-synaptic neuron** and the second neuron is **the post-synaptic neuron**. A small gap between the pre and postsynaptic membranes is called **Synaptic Cleft** that forms a structural gap and a functional bridge between neurons. The axon terminals contain synaptic vesicles filled with **neurotransmitters**. When an impulse [action potential] arrives at the axon terminals, it depolarizes the pre-synaptic membrane, opening the voltage gated calcium channels. Influx of calcium ions stimulates the synaptic vesicles towards the pre-synaptic membrane and fuses with it. In the neurilemma, the vesicles release their neurotransmitters into the synaptic cleft by **exocytosis.** The released neurotransmitters bind to their specific receptors on the post-synaptic membrane, responding to chemical signals. The entry of the ions can generate a new potential in the post-synaptic neuron, which may be either **excitatory** or **inhibitory**. Excitatory post-synaptic potential causes depolarization whereas inhibitory post-synaptic potential causes **hyperpolarization** of post-synaptic membrane (Figure 10.6).

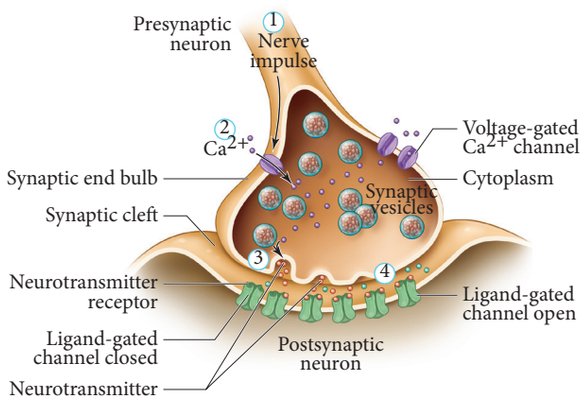


Figure 10.6 Synaptic Transmission

10.4 Central neural system (CNS)

The CNS includes the brain and the spinal cord, which are protected by the bones of the skull and vertebral column. During its embryonic development, CNS develops from the ectoderm.

Can you state why some areas of the brain and spinal cord are grey and some are white?

10.4.1 Brain

The brain acts as the command and control system. It is the site of information

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